

FORESTBURG AGCM, AB, Canada

WMO: 716740

Lat: 52.5485N

Lon: 112.1234W

Elev: 721

StdP: 92.96

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -30.7	(c) -27.8	(d) -34.3	(e) 0.2	(f) -30.5	(g) -31.2	(h) 0.2	(i) -27.7	(j) 11.4	(k) -1.3	(l) 10.7	(m) 0.6	(n) 2.4	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	29.1	17.5	27.2	16.5	25.5	15.8	19.5	26.2	18.2	24.8	17.2	23.3	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	13.2	23.0	15.8	12.2	21.4	14.7	11.4	20.0	58.7	26.2	54.6	25.3	51.0	23.2	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.4	8.9	7.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-35.0	32.3	2.5	2.4	-36.8	34.0	-38.3	35.3	-39.7	36.7	-41.5	38.4
			-35.1	21.7	2.5	1.4	-36.9	22.6	-38.4	23.4	-39.7	24.2	-41.5	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-9.8	-10.9	-4.9	3.1	10.8	14.9	17.0	15.9	10.9	4.3	-4.3	-11.0
	DBStd	11.98	8.28	8.57	7.90	5.33	4.27	2.87	2.87	3.04	4.42	4.92	7.12	7.58
	HDD10.0	3222	612	586	461	212	42	2	0	1	41	185	429	652
	HDD18.3	5603	871	819	719	458	234	108	60	87	223	434	679	911
	CDD10.0	701	0	0	0	5	67	149	218	182	69	9	1	0
	CDD18.3	39	0	0	0	0	1	6	20	11	2	0	0	0
	CDH23.3	943	0	0	0	4	84	116	349	286	99	3	0	0
	CDH26.7	219	0	0	0	0	13	16	96	69	24	0	0	0
Wind	WSAvg	3.8	3.9	3.8	3.6	4.1	3.9	4.0	3.3	3.3	3.4	4.3	3.7	3.8
Precipitation	PrecAvg	412	15	12	16	18	42	80	78	59	44	16	12	16
	PrecMax	514	42	35	45	46	120	157	153	145	120	42	30	44
	PrecMin	319	3	0	1	1	10	24	23	9	0	1	0	2
	PrecStd	61	10	9	11	13	27	42	36	36	31	11	8	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.4	13.4	22.7	28.3	28.4	31.6	30.9	29.8	23.3	14.4	6.6
		MCWB	3.2	3.6	7.0	9.7	13.2	16.4	19.1	17.1	15.5	12.5	7.3	3.4
	2%	DB	5.2	4.7	10.1	18.4	25.8	26.4	29.2	28.6	26.1	18.5	9.9	4.1
		MCWB	2.1	1.6	5.3	8.2	12.6	16.0	19.2	17.0	14.1	10.0	5.0	1.3
	5%	DB	3.5	2.8	6.6	15.1	23.1	24.4	27.2	26.5	23.5	15.4	6.7	1.7
		MCWB	1.1	0.4	3.0	6.8	11.6	15.3	18.1	16.5	13.4	8.3	2.9	-0.7
	10%	DB	1.7	1.0	4.4	12.3	20.2	22.3	25.0	24.5	20.5	12.5	4.7	-0.4
		MCWB	-0.5	-0.8	1.5	5.6	10.3	14.2	17.4	16.0	12.3	6.8	1.6	-2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	3.6	8.1	10.7	14.5	19.0	21.9	20.1	16.6	13.1	8.1	3.6
		MCDB	6.3	7.0	13.2	22.1	25.4	25.6	28.1	26.3	25.9	21.8	14.8	6.4
	2%	WB	2.3	1.8	5.1	8.6	13.2	17.4	20.1	18.7	15.0	10.8	5.3	1.3
		MCDB	4.8	4.3	9.4	17.4	23.1	23.9	27.1	25.7	24.1	16.4	8.9	3.9
	5%	WB	1.2	0.5	3.2	7.2	12.2	16.2	18.9	17.5	13.7	8.9	3.3	-0.5
		MCDB	3.6	2.5	6.2	14.2	21.2	22.1	25.4	24.4	21.8	14.7	6.2	1.6
	10%	WB	-0.4	-0.7	1.7	5.9	11.2	15.0	17.8	16.4	12.7	7.3	1.9	-2.3
		MCDB	1.6	1.1	4.2	11.7	19.1	20.5	24.0	22.9	20.0	12.2	4.1	-0.5
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.5	10.2	11.9	14.0	12.2	13.5	14.2	14.0	12.3	9.9	9.2
		MCDBR	10.2	11.8	12.5	17.8	18.6	15.8	17.5	18.8	20.2	17.2	12.4	10.6
		MCWBR	7.9	9.3	8.3	9.2	8.0	7.1	8.6	8.8	9.3	9.6	8.2	8.4
	5% WB	MCDBR	10.5	11.0	11.7	16.1	16.4	13.3	15.7	16.5	18.1	16.1	11.6	10.6
		MCWBR	8.4	8.8	8.1	8.6	7.4	6.8	8.4	8.4	8.7	9.2	8.1	8.6
Clear-Sky Solar Irradiance	taub		0.267	0.283	0.293	0.331	0.352	0.364	0.358	0.372	0.317	0.284	0.258	0.247
	taud		2.276	2.320	2.413	2.361	2.351	2.350	2.400	2.341	2.460	2.530	2.457	2.319
	Ebn at Noon		731	837	907	905	898	887	886	851	869	839	762	699
	Edh at Noon		62	80	90	109	116	117	110	110	86	65	51	49
All-Sky Solar Radiation	RadAvg		1.08	2.13	3.57	4.73	5.76	6.03	6.33	5.17	3.61	2.15	1.09	0.80
	RadStd		0.10	0.17	0.26	0.45	0.37	0.40	0.32	0.38	0.38	0.27	0.11	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page